```
SELECT *
FROM `retail-sales-project-494107.retail_dataset. Retail_sales_table` LIMIT 10;
```

Output:

Row	Transaction ID	Date	Customer ID	Gender
1	191	2023-10-18	CUST191	Male
2	204	2023-09-28	CUST204	Male
3	230	2023-04-23	CUST230	Male
4	232	2023-02-06	CUST232	Female
5	309	2023-12-23	CUST309	Female
6	310	2023-10-12	CUST310	Female
7	363	2023-06-03	CUST363	Male
8	371	2023-02-21	CUST371	Female
9	397	2023-03-10	CUST397	Female
10	454	2023-02-22	CUST454	Female

```
---Question 1: WHERE Clause
---Filter all transactions that occurred in the year 2023.
---Expected Output: All columns

SELECT *
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`
WHERE Date BETWEEN '2023-01-01' AND '2023-12-31';
```

Output: 1st 9

Row	Transaction ID	Date	Customer ID	Gender	Age
1	191	2023-10-18	CUST191	Male	
2	204	2023-09-28	CUST204	Male	
3	230	2023-04-23	CUST230	Male	
4	232	2023-02-06	CUST232	Female	
5	309	2023-12-23	CUST309	Female	
6	310	2023-10-12	CUST310	Female	
7	363	2023-06-03	CUST363	Male	
8	371	2023-02-21	CUST371	Female	
9	397	2023-03-10	CUST397	Female	

```
---Question 2: Filtering + Conditions
```

```
---Display all transactions where the Total Amount is more than the average Total Amount of
the entire dataset.
---Expected Output: All columns

SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`;
```

Output:

Query results

Job information | Results

Row	Total_Revenue
1	456000

```
---Question 3: Aggregate Functions
--Calculate the total revenue (sum of Total Amount).
---Expected Output: Total_Revenue

SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`;
```

Output:

Row	Total_Revenue
1	456000

```
---Question 4: DISTINCT
---Display all distinct Product Categories in the dataset.
---Expected Output: Product_Category
SELECT DISTINCT `Product Category`
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`;
```

Output:

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

```
---Question 5 GROUP BY
```

```sql
---For each Product Category, calculate the total quantity sold.
---Expected Output: Product_Category, Total_Quantity
SELECT `Product Category`,
SUM(Quantity) AS Total_Quantity
FROM`retail-sales-project-494107.retail_dataset.Retail_sales_table`
GROUP BY `Product Category`;
```

Output:

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

```sql
---Question 6: CASE Statement
---Create a column called Age_Group that classifies customers as:
□ 'Youth' (<30)
□ 'Adult' (30□59)
□ 'Senior' (60+)
---Expected Output: Customer_ID, Age, Age_Group

SELECT 'Customer ID', Age,
CASE
  WHEN Age <30 THEN 'Youth'
  WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
  ELSE 'Senior'
  END AS Age_Group
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`;
```

Output:1st 15

Row	f0_	Age	Age_Group
1	Customer ID	64	Senior
2	Customer ID	39	Adult
3	Customer ID	54	Adult
4	Customer ID	43	Adult
5	Customer ID	26	Youth
6	Customer ID	28	Youth
7	Customer ID	64	Senior
8	Customer ID	20	Youth

9	Customer ID	30	Adult
10	Customer ID	46	Adult
11	Customer ID	57	Adult
12	Customer ID	36	Adult
13	Customer ID	51	Adult
14	Customer ID	46	Adult
15	Customer ID	54	Adult

```
---Question 7: Conditional Aggregation
---For each Gender, count how many high-value transactions occurred (where Total Amount > 500).
---Expected Output: Gender, High_Value_Transactions

SELECT Gender,COUNT('Transaction ID') AS High_Value_Transactions
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`
WHERE `Total Amount` >500
GROUP BY Gender;
```

Output:

Row	Gender	High_Value_Tran...
1	Female	155
2	Male	144

```
---Question 8 HAVING Clause
---For each Product Category, show only those categories where the total revenue exceeds 5,000.
---Expected Output: Product_Category, Total_Revenue
SELECT `Product Category`, SUM(`Total Amount`) AS Total_Revenue
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) >5000;
```

Output:

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

```
---Question 9: Calculated Fields
---Display a new column called Unit_Cost_Category that labels a transaction as:
--- 'Cheap' if Price per Unit < 50
---'Moderate' if Price per Unit between 50 and 200
--'Expensive' if Price per Unit > 200
---Expected Output: Transaction_ID, Price_per_Unit, Unit_Cost_Category

SELECT Transaction ID, Price per unit,
CASE
  WHEN Price per Unit< 50 THEN 'Cheap'
   WHEN `Price per Unit`BETWEEN 50 AND 200 THEN 'Moderate'
   WHEN Price per Unit> 200 THEN 'Expensive'
   END AS Unit_Cost Category
FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`;
```

Output:

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

```
---Question 10: Combining WHERE + CASE
---Display all transactions from customers aged 40 or older and add a column Spending_Level
---showing:
- 'High' if Total Amount > 1000
- 'Low' otherwise
---Expected Output: Customer_ID, Age, Total_Amount, Spending_Level

SELECT `Customer ID`, Age, `Total Amount`,
CASE
  WHEN `Total Amount` >1000 THEN 'High'
  ELSE 'Low'
  END AS Spending_Level
  FROM `retail-sales-project-494107.retail_dataset.Retail_sales_table`
  WHERE Age >=40;
```

Output:1st 10

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low
9	CUST855	54	25	Low
10	CUST967	62	25	Low